ROC AUC CALCULATION

The ROC curve and Area Under the Curve (AUC) value are used to measure the ability of the model to separate classes. A value of 0.5 indicates that the model is as accurate as random chance and a value of 1.0 represents perfect discrimination. To create the ROC curve in WEKA, I right-clicked on the result in the Result list and selected Visualize threshold curve. I plotted False Positive Rate on X axis, and True Positive Rate on Y axis.

Table 9 : ROC AUC Scores by Performance Class.

Class	ROC AUC	Interpretation
(-inf-65.5] Low	0.870	Good — well above 0.5 random baseline
(65.5-68.5] Medium	0.696	Acceptable — harder class due to narrow score range
(68.5-inf) High	0.858	Good — well above 0.5 random baseline
Weighted Average	0.807	Strong discriminative ability overall

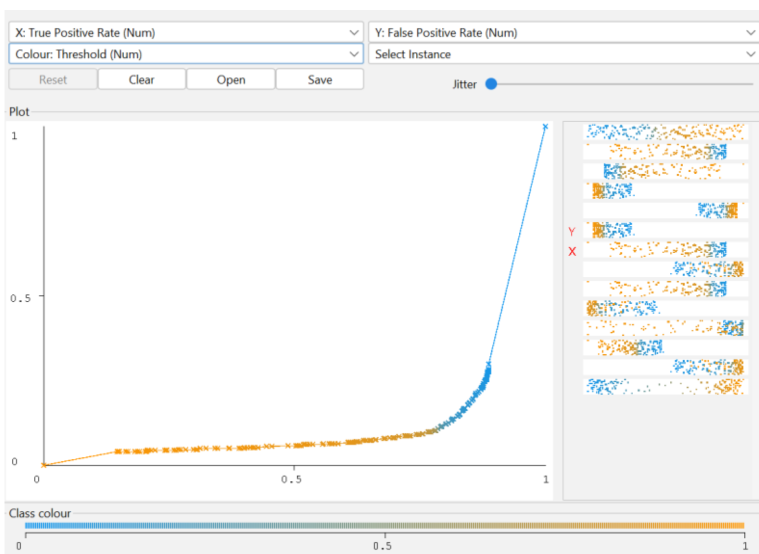

Figure 51 : WEKA Threshold Curve Visualizer: ROC Curve of J48. Area Under ROC = 0.8701. The high slope towards the top-left corner demonstrates that the discriminative ability is very high and far above the random classifier baseline of 0.5.